

Education

Computer Science Ph.D. Student

Research area: Human Computer Interaction

Advisor: **Zhu-Tian Chen**

University of Minnesota, Twin Cities

Aug 2024 – Now

B.Eng. with Honors in Artificial Intelligence (Turing Class)

Outstanding honored undergraduate in **Chu Kochen Honors College**

President of the Morningside Scholars Program in class 2023

Zhejiang University

Sep 2020 – June 2024

Research Interest & Coursework

Research interest

AR/VR/HCI/Data Visualization/AI x XR

Coursework

Machine Learning/Computer Vision/Natural Language Process/Data Visualization
Programming in Virtual Reality/Information and Interactive Design

Research Experience

Harvard Visual Computing Group, Harvard University

Research Intern with **Zhu-Tian Chen** and **Hanspeter Pfister**

Cambridge, USA

Jan 2024 - May 2024

- Develop a 3D immersive data analysis system enabling in-situ and ex-situ combined workflow
- WebXR platform oriented design, combining hand gesture interaction into the immersive analytics workflow
- Design new data analysis paradigm for analyzing and viewing 2D/3D, static/dynamic, tangible/virtual data

Stanford HCI Group, Stanford University

Research Intern with **Jackie Yang** and **James Landay**

California, USA

Jul 2023 - Sep 2023

- **SwiftGenieXR** — a declarative programming framework, exploring generative AI in XR workspace
- Implement a Swift-based framework facilitating voice/hand input and LLM interaction in Apple Vision Pro
- Harness SwiftMacro and SwiftData to tackle environment understanding and persistent data storage problem

ZJU International Design Institute, Zhejiang University

Student Research Training Program advised by **Lingyun Sun** and **Zejian Li**

Hangzhou, China

Mar 2022 - Jun 2023

- Design non-parametric algorithms to improve the sampling efficiency in DDIM/DDPM diffusion models
- Implement the proposed algorithm in baseline diffusion models and deploy them on training
- Collect and organize experimental image dataset, analyze and visualize the experiment result

Project

Handbot——Apple WWDC2023 Swift Student Challenge

[Github] [Youtube]

Human-centered AI

Apr 2023

- A swift playground aiming to encourage users to think critically, and recreate all the AIGC content.
- Concentrate the design paradigm on three information modals — text, image and audio.
- First solo work in collaboration with ChatGPT and learn a lot about human AI interaction

Transciscense——Immersive Game Watching App For The Hearing-impaired

[Github] [Youtube]

Accessibility Design

Jun 2023

- A multimodal-designed swift app for the hearing-impaired users to immerse in watching sports game.
- Utilize the haptic and visual information-color, shape, vibration to compensate for the listening channel
- Design a data visualization of the game watching experience for better review and summary

Is raining threes worth it?——NBA 3-pointer Data Visualization Project

[Website] [Youtube]

Data Visualization

Dec 2022

- A front-end interactive visualization website displaying the NBA 3-pointers data, received 17000+ views
- Analyze the impact of the three-point shot on the game and the player from multiple perspectives.
- This project enlightens me to transition from 2D visualization to immersive 3D visualization

Cut_Fruits_3D——immersive Fruit Ninja VR Game

[Github] [Youtube]

Virtual Reality

Jun 2023

- An immersive 3D version Fruit Ninja VR Game, received 500+ views, developed by Unity3D
- Support object collision detection and vibration feedback in controller, along with sound effect feedback
- Encourage user to stretch their body and move around, thus good for player's health

Professional skills

Programming Tools Language	Unity3D/Python/C/C++/C#/Swift/WebXR/HTML/CSS/Javascript/Pytorch/Bash Apple Vision Pro/Meta Quest/Generative AI/Figma/Xcode/Swift Playground/Pr/PS English/Chinese/Mandarin/Cantonese/a little bit French and Japanese
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Teaching Experience

UMN Spring CSCI5115 User Interface Design and Evaluation <i>Instructor: Loren Terveen</i>	Teaching Assistant <i>Jan 2025 - present</i>
• Provide instructions for qualitative and quantitative user study and evaluation, formative Study and problem conceptualization in human-centered design and product prototype	
Hangzhou Yungu Primary School Swift CCA program <i>SwiftUI teaching</i>	Teacher <i>October 2023 - present</i>
• Teach SwiftUI and Swift programming knowledge to three primary school students	

Extracurricular activities

Morningside Cultural China Scholarship Program <i>social science and Chinese traditional culture</i>	President <i>May 2023 - Dec 2023</i>
• A program fosters deep understanding of Chinese traditional culture as well as leadership • Provide chance to go on academic visit to USA, Europe, Singapore, Hongkong and Macau	
Lanshiren Mountain and Forest hiking club <i>sports and environmental protection</i>	Leader <i>Sep 2022 - Jun 2023</i>
• Non-profit organization arouses people's interest into hiking and protecting the natural environment • This club will organize picking rubbish and cleaning up the natural environment at regular intervals	
Kuafu running and jogging club <i>sports</i>	Leader <i>Sep 2023 - Dec 2023</i>
• Campus association, which encourages running and long-distance running(e.g. marathon) for exercise	
Sichuan and Chongqing intangible cultural heritage summer social practice <i>social practice and community commitment</i>	Member <i>Aug 2022</i>
• Fieldwork records protect and inherit China's intangible cultural heritage in southwest China's provinces	
Inner Mongolia poverty alleviation education summer social practice <i>social practice and community commitment</i>	Member <i>Aug 2021</i>
• Concentrate on the economy and education situation of the less developed inland areas of China	
Youtube and Bilibili platform vlogger <i>Media art and content creation</i>	Creator <i>Aug 2020 - present</i>
• Passionate about spreading knowledge and what I've learned to help others on streaming platforms	

Honors

Second-prize Scholarship of Zhejiang University	Oct 2023
2022 Hangzhou Olympic Asian Games Conditional Volunteer	May 2023
Top 5 course projects of Introduction to Data Visualization	Jan 2023
Chu Ko Chen College summer social practice second prize	Sep 2022

Hobbies

Sports Arts	swimming/basketball/hiking/jogging/cycling/ping-pong photography/vlog/podcast/saxophone/movie
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